

SOGO-HAYA, Russian Federation

WMO: 245570

Lat: 64.283N

Lon: 126.450E

Elev: 85

StdP: 100.31

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -49.6	(c) -48.0	(d) -51.5	(e) 0.0	(f) -49.4	(g) -49.8	(h) 0.0	(i) -48.1	(j) 8.0	(k) -31.4	(l) 7.2	(m) -33.5	(n) 1.0	(o) 110	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	28.7	19.2	26.6	18.2	24.5	17.0	20.5	26.6	19.2	24.4	17.9	23.0	2.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.4	23.0	17.1	12.3	22.2	15.8	11.3	20.7	59.8	26.3	55.2	24.3	51.1	23.0	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.7	(b) 6.8	(c) 6.2	(e) -50.9	(f) 31.9	(g) 2.3	(h) 1.8	(i) -52.5	(j) 33.2	(k) -53.8	(l) 34.3	(m) -55.1	(n) 35.3	(o) -56.7	(p) 36.7		
(a) 7.7	(b) 6.8	(c) 6.2	(e) -50.9	(f) 22.6	(g) 2.3	(h) 2.0	(i) -52.5	(j) 24.1	(k) -53.8	(l) 25.2	(m) -55.1	(n) 26.3	(o) -56.7	(p) 27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.8	-38.3	-31.6	-17.8	-5.6	5.3	15.0	17.9	14.6	5.7	-7.9	-28.0	-36.1		
	DBStd	20.90	6.37	6.02	7.80	6.15	5.46	4.27	4.05	3.72	4.43	5.99	6.92	6.97		
	HDD10.0	7440	1498	1165	862	468	166	8	1	6	141	555	1140	1429		
	HDD18.3	9967	1757	1398	1121	718	404	116	58	126	378	813	1390	1688		
	CDD10.0	583	0	0	0	0	21	157	244	147	13	0	0	0		
	CDD18.3	68	0	0	0	0	0	16	44	9	0	0	0	0		
	CDH23.3	693	0	0	0	0	3	192	419	78	1	0	0	0		
	CDH26.7	174	0	0	0	0	0	40	126	8	0	0	0	0		
	WSAvg	2.5	2.2	2.6	2.6	2.9	2.6	2.5	2.7	2.5	2.4	2.7	2.4	2.3		
	PrecAvg	364	13	11	13	17	35	42	53	53	45	43	24	14		
(15) Precipitation	PrecMax	522	28	30	35	38	70	92	108	129	97	79	57	31		
	PrecMin	257	5	0	2	5	5	13	17	13	4	6	9	5		
	PrecStd	54	6	7	8	8	16	19	23	23	23	16	11	6		
	PrecStd	54	6	7	8	8	16	19	23	23	23	16	11	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-19.5	-14.8	1.2	9.5	23.1	29.8	32.4	27.7	21.0	4.8	-7.8	-18.8		
		MCWB	-19.8	-15.4	-1.3	4.1	13.5	18.2	21.8	18.5	14.2	3.5	-9.2	-19.5		
	2%	DB	-23.5	-16.7	-1.5	7.1	19.9	27.8	29.8	25.4	17.7	2.9	-11.4	-21.6		
		MCWB	-23.8	-17.2	-3.4	2.1	10.7	18.0	20.3	17.6	12.1	1.6	-11.9	-22.0		
	5%	DB	-26.9	-20.5	-3.6	5.4	17.1	25.5	27.7	23.3	14.8	1.2	-14.3	-23.3		
		MCWB	-27.1	-20.9	-5.4	1.0	9.7	16.8	19.2	16.9	10.2	0.0	-14.7	-23.6		
	10%	DB	-29.5	-23.0	-5.6	3.7	14.3	23.3	25.5	21.2	12.5	-0.4	-18.3	-25.9		
		MCWB	-29.7	-23.3	-7.0	0.0	8.1	15.9	18.1	16.0	8.7	-1.3	-18.6	-26.1		
	0.4%	WB	-19.8	-15.3	-0.7	4.7	13.6	20.9	22.8	20.5	15.6	3.9	-9.0	-19.3		
		MCDB	-19.4	-15.0	1.1	8.8	21.3	27.4	29.8	25.5	20.4	4.3	-8.1	-18.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-23.7	-17.3	-3.3	2.7	11.5	18.7	21.4	18.7	12.5	2.1	-11.9	-21.9		
		MCDB	-23.3	-16.8	-1.7	6.1	18.4	25.3	28.0	22.8	16.1	2.8	-11.4	-21.6		
	5%	WB	-27.0	-20.9	-5.3	1.5	10.0	17.5	20.2	17.6	11.1	0.2	-14.7	-23.5		
		MCDB	-26.9	-20.5	-3.8	4.6	16.6	24.1	26.1	21.8	14.3	1.0	-14.3	-23.2		
	10%	WB	-29.7	-23.4	-7.1	0.3	8.3	16.4	19.0	16.5	9.3	-1.2	-18.5	-26.1		
		MCDB	-29.5	-23.1	-5.6	3.6	13.8	22.1	24.1	20.6	11.9	-0.4	-18.2	-25.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	7.2	9.9	10.6	9.2	10.5	9.8	9.5	8.1	5.1	5.5	5.0		
		MCDBR	6.5	7.5	10.5	10.7	12.5	14.5	12.9	12.9	11.8	5.4	5.8	7.0		
		MCWBR	6.4	7.2	8.8	7.0	6.9	6.8	5.5	6.2	6.6	4.0	5.7	6.9		
		MCDBR	6.5	7.7	10.2	9.5	12.0	13.6	11.3	10.9	11.1	4.7	5.7	6.8		
	5% WB	MCWBR	6.5	7.4	8.7	6.5	6.8	7.0	5.7	5.7	6.7	3.8	5.7	6.7		
		MCWBR	6.5	7.4	8.7	6.5	6.8	7.0	5.7	5.7	6.7	3.8	5.7	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.189	0.245	0.290	0.338	0.387	0.407	0.485	0.469	0.337	0.266	0.187	0.179		
	taud		1.902	2.082	2.091	2.090	2.221	2.292	2.044	2.035	2.406	2.258	1.910	1.886		
	Ebn at Noon		368	665	784	824	816	808	715	680	744	641	352	140		
	Edh at Noon		27	62	97	123	120	115	144	132	73	52	25	11		
(44) All-Sky Solar Radiation	RadAvg		0.19	1.06	2.90	4.98	5.39	5.88	5.36	3.96	2.36	0.94	0.33	0.07		
	RadStd		0.01	0.05	0.14	0.21	0.49	0.41	0.27	0.35	0.33	0.12	0.03	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	-162	N/A	N/A

Nomenclature: See separate page